

Program Name: A program print data type in Java

Code:

```
public class dataType {
    public static void main(String[] args){
        byte a = 10;
        short b = 100;
        int c = 100000;
        char d = 'A';
        float e = 3.50f;
        double f = 3.4567;

        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
        System.out.println(e);
        System.out.println(f);
    }
}
```

Output:

```
C:\Users\abrar\.jdk\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
10
100
100000
A
3.5
3.4567

Process finished with exit code 0
```

Program Name: A program to access array elements

Code:

```
public class accessArrayElement {
    public static void main(String[] args){
        int[] age = {16, 23, 45};
        System.out.println("Accessing array elements:");
        System.out.println("First Element: " + age[0]);
        System.out.println("Second Element: " + age[1]);
        System.out.println("Third Element: " + age[2]);
    }
}
```

Output:

```
C:\Users\abrar\.jdk\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
Accessing array elements:
First Element: 16
Second Element: 23
Third Element: 45

Process finished with exit code 0
```

Program Name: A program to implement for loop

Code:

```
public class forLoopArray {  
    public static void main(String[] args){  
        int[] a = {12,23,34,45,56};  
  
        for(int i=0; i<a.length; i++){  
            System.out.println("index[" + i + "] = " + a[i]);  
        }  
    }  
}
```

Output:

```
C:\Users\abrar\.jdk\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ  
index[0] = 12  
index[1] = 23  
index[2] = 34  
index[3] = 45  
index[4] = 56  
  
Process finished with exit code 0
```

Program Name: A program to implement for each loop

Code:

```
public class forEachLoopArray {  
    public static void main(String[] args){  
        int[] a = {10,20,30,40,50};  
        int i = 0;  
        for(int x : a){  
            System.out.println("index[" + i + "] = " + x);  
            i++;  
        }  
    }  
}
```

Output:

```
C:\Users\abrar\.jdk\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ  
index[0] = 10  
index[1] = 20  
index[2] = 30  
index[3] = 40  
index[4] = 50  
  
Process finished with exit code 0
```

Program Name: A program to search an element in array

Code:

```
public class linearSearch {
    public static void main(String[] args){
        int[] a = {10,20,30,40,50};
        int item = 40;
        int count = 0;
        boolean t = true;

        for(int x : a){
            if(x == item){
                t = false;
                break;
            }
        }
        System.out.println((t) ? "Item Not Found :(" : "Item Found :)");
    }
}
```

Output:

```
C:\Users\abrar\.jdk\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
Item Found :)
```

```
Process finished with exit code 0
```

Program Name: A program to find the maximum value in array

Code:

```
public class maxElement {
    public static void main(String[] args){
        int[] a = {10,20,30,40,50};
        int max = a[0];

        for(int i=1; i<a.length; i++){
            if(max < a[i]){
                max = a[i];
            }
        }
        System.out.println("Maximum Element of the Array: " + max);
    }
}
```

Output:

```
C:\Users\abrar\.jdk\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
Maximum Element of the Array: 50
```

```
Process finished with exit code 0
```